Data Engineering Integration 10.4.0: Security for Azure Databricks

In this article, learn the security functionalities that Data Engineering Integration supports and how to configure them in a Microsoft Azure Databricks environment.

Overview

Data Engineering Integration can connect to a Databricks cluster on Azure. Databricks incorporates the open-source Apache Spark cluster technologies and capabilities.

The integration of the Informatica domain with the Databricks environment requires Informatica connection objects and cluster configurations. A cluster configuration is a domain object that contains configuration parameters that you import from the Databricks cluster. You then associate the cluster configuration with connections to access the Databricks environment.

To integrate the domain with a secure Databricks cluster, it is not mandatory to configure security on the Informatica domain. The integration leverages the built-in token-based security provided by the Azure platform.

This article discusses Data Engineering Integration support for security mechanisms that an Azure Databricks cluster uses.

Supported Security Systems for Azure Databricks Clusters

The Data Integration Service uses token authentication to access Databricks.

You can apply the following security types to data in the Azure Databricks environment:

- **Resources on ADLS** are secured by OAUTH credentials.
- **Resources on WASB** are secured by an account access key.

Note: Delta Lake tables reside on ADLS storage and are not supported on WASB storage.

Documentation Reference

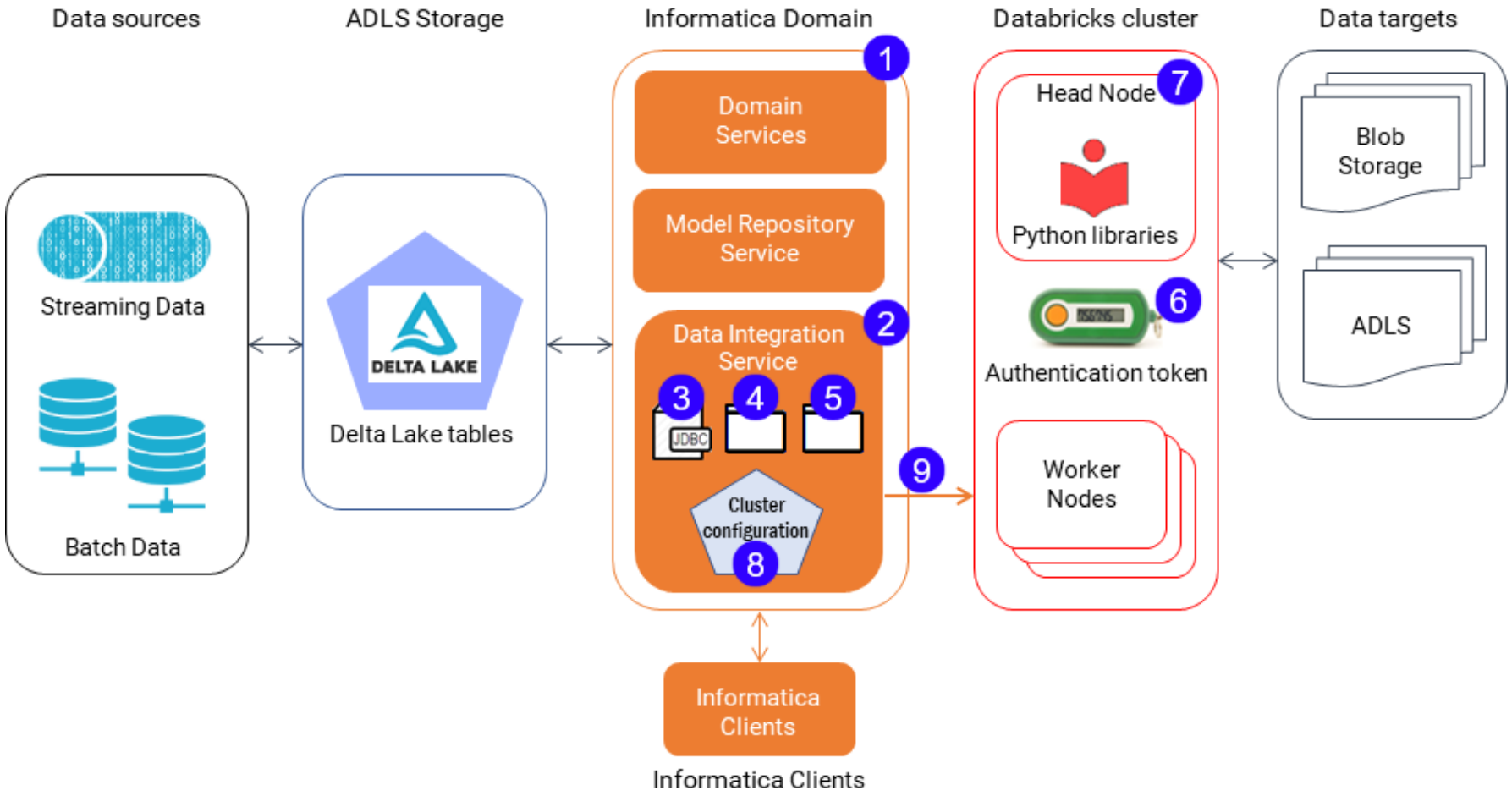
For more information, refer to the following documentation:

- Accessing Azure blob storage from Databricks clusters
- Using Delta Lake storage with Databricks
- Accessing ADLS Gen1 and ADLS Gen2 storage from Databricks clusters
- Integrating Databricks with Informatica Data Engineering Integration

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Security Configuration for Azure Databricks

The following image shows the security architecture of a Data Engineering Integration integration with an Databricks cluster that uses Azure Active Connect and Azure Active Directory.



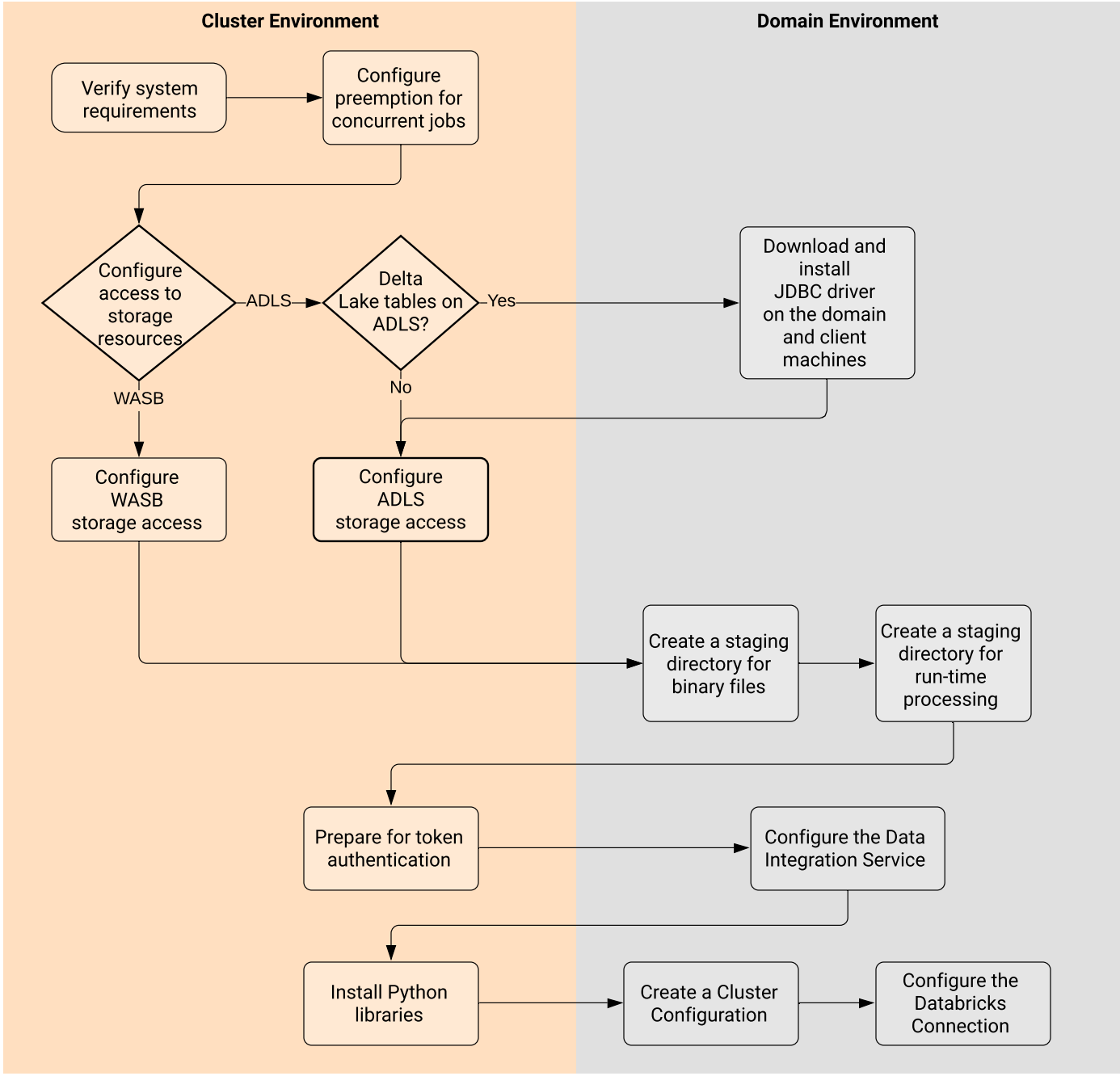
The numbered elements in the image are explained in the following key.
Note: Not all elements are mandatory, depending on your environment. See the flowchart on the next page for more information.

1. The Informatica domain is deployed in the Azure environment.
2. The Data Integration Service manages mapping jobs and sends jobs to the Databricks cluster for processing.
3. The JDBC driver enables the domain to connect to Delta Lake tables.
4. Create a staging directory for binary files.
5. Create a staging directory for run-time processing.
6. Generate an authentication token on the Databricks cluster and provide it to the Data Integration Service.
7. Optionally import additional Python libraries.
8. Import the cluster configuration from the Databricks cluster.
9. The Databricks connection enables communication between the domain and the Databricks cluster.

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Security Configuration for Azure Databricks

The following flowchart shows the Data Engineering Integration security configuration tasks and where to perform the tasks. Click on the task to see the topic in documentation.



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